

LEFTMOST REASSOCIATION OF DYCK COMPONENTS: EXACT TRANSIENT LAYERS AND TERMINAL DEPTH FIBRES

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ABSTRACT. Let a nonempty Dyck path have primitive factorisation $P = C_1 \cdots C_k$, with $C_1 = UAD$. We iterate the deterministic rule that fixes P when $k = 1$ and otherwise replaces its first two components by UAC_2D . The complete orbit is explicit: after t updates, the first component has absorbed $C_2, \dots, C_{t+1}$. Consequently the entry time is exactly $k - 1$, all recurrent paths are fixed and primitive, and the unique path of maximum depth $n - 1$ in semilength n is $(UD)^n$. The number of paths at depth $k - 1$ is

$$\frac{k}{2n-k} \binom{2n-k}{n}.$$

We also resolve every terminal basin by depth. If a fixed target is UQD and the interior Q has r primitive factors, then it has exactly one basin state at each depth $0, \dots, r$. Hence its depth-fibre polynomial is $1 + u + \cdots + u^r$, and the unique largest terminal fibre has size n . First-return decomposition, Catalan and ballot enumeration, ground-level comb/Tamari covers, deterministic leftmost-rotation precedents, and the ordered-tree graft/lift representation are treated as background inputs. The residual is only the exact temporal/target-fibre conjunction for this literal selector. No priority follows, and external circulation remains on hold.

1. INTRODUCTION

A local cover relation does not by itself determine a dynamical system: one must also specify which available cover is chosen at each state. This note fixes a particularly rigid selector on Dyck paths. At every nonfixed state, we perform the reassociation at the leftmost return to ground level. The selected down-step crosses the next primitive Dyck component, and no other part of the path changes.

The atomic reassociation belongs to the Tamari setting. Associativity orders and their lattice property go back to Huang and Tamari [3]; path formulations of Tamari-type covers also appear in the m -Tamari literature [1]. Primitive decomposition of Dyck words and enumeration by the number of primitive components are likewise established topics [6], while Catalan and ballot enumeration are standard background [7]. We assign all of these ingredients zero contribution credit.

Two closer rotation precedents further narrow the boundary. Pallo studies a uniquely selected leftmost left-rotation on binary trees and the resulting rooted directed tree, rank, and distance [5, pp. 802–803]. That deterministic map is not the map below: after adjoining a loop at its terminal root it has one fixed tree, whereas our map has Cat_{n-1} fixed paths, more than one for $n \geq 3$. Thus it is neither equal nor conjugate, by mirror or reversal, to our map. Separately, Chapoton identifies the comb-order covers on Dyck paths as exactly the Tamari covers whose moved subpath lies at height zero [2, §1.2, p. 438], tracing the comb order to Pallo’s right-arm rotations [4]. Every nonfixed update below is one of those ground-level covers, chosen at the leftmost ground return. The idea of a deterministic leftmost scheduler and the ground-level cover correspondence both receive zero contribution credit.

The purpose here is narrower: to record, for this particular selector, the exact conjunction of its full temporal law with its target-indexed inverse fibres. None of the scheduler, its individual covers, its tree model, or its component census is claimed separately. The argument has four parts.

- (1) We give a closed formula for every iterate. It turns the number of primitive factors into the pointwise clock $\tau(P) = \kappa(P) - 1$ and identifies $(UD)^n$ as the unique depth- $n - 1$ path.
- (2) We derive the complete temporal census: depth $k - 1$ contains $\frac{k}{2n-k} \binom{2n-k}{n}$ states.
- (3) For every fixed target and every feasible depth, we construct one basin state and prove that no second state exists at that depth.
- (4) We derive the depth-fibre polynomial and classify the unique target of largest fibre.

The closed iterate in [lemma 3.1](#) is the common mechanism: it proves the clock and leaves only a suffix cut to invert an endpoint. No figure is needed because that formula exposes the whole state change.

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This package is owner-thin. The source boundary in [section 7](#) is a conservative credit assignment, not a priority or originality decision. The draft is maintained for internal theorem evaluation under `HOLD_EXTERNAL`.

2. DYCK FACTORS AND THE LITERAL MAP

A *Dyck path of semilength n* is a word in U, D with n occurrences of each letter such that every prefix contains at least as many U 's as D 's. Let $\mathcal{D}_n$ denote the set of such paths. A nonempty Dyck path is *primitive* if its only return to height zero occurs after its final step. Cutting at all positive returns gives a unique factorisation

$$(1) \quad P = C_1 C_2 \cdots C_k$$

into primitive Dyck paths. We write $\kappa(P) = k$.

Every primitive first factor has a unique first-return form $C_1 = UAD$ with A a possibly empty Dyck path. Define

$$(2) \quad \Phi_n(P) = \begin{cases} P, & k = 1, \\ UAC_2 D C_3 \cdots C_k, & k \geq 2. \end{cases}$$

We suppress the subscript n when the semilength is fixed. The word AC_2 is a Dyck path, so $UAC_2 D$ is primitive. Thus (2) lies in $\mathcal{D}_n$ and is well defined.

There is a standard ordered-tree form of the same rule. Under the contour bijection between Dyck paths and rooted ordered plane trees [7, Theorem 1.5.1], the primitive factors $C_1, \dots, C_k$ encode the ordered subtrees rooted at the k children of the tree root. Write these subtrees as $T_1, \dots, T_k$. For $k \geq 2$, (2) is precisely

$$(3) \quad (T_1, T_2, T_3, \dots, T_k) \mapsto (T_1 \triangleleft T_2, T_3, \dots, T_k),$$

where $T_1 \triangleleft T_2$ appends T_2 as the rightmost child of the root of T_1 . Thus the factor clock below is just root degree minus one. This contour conjugacy and graft description are representation-level background, not residual claims.

Let $\tau(P)$ be the least $t \geq 0$ for which $\Phi^t(P)$ is fixed, and put

$$E(P) = \Phi^{\tau(P)}(P).$$

A path is *recurrent* when it lies on a directed cycle of Φ_n . For a fixed target T , define its depth-fibre polynomial

$$(4) \quad \mathcal{B}_T(u) = \sum_{P: E(P)=T} u^{\tau(P)}.$$

The complete claim package follows. We use $\text{Cat}_m = \frac{1}{m+1} \binom{2m}{m}$.

Theorem 2.1 (Clock, layers, and terminal depth fibres). *For every $n \geq 1$, the map Φ_n has the following properties.*

- (i) *Every nonfixed update reduces κ by exactly one. The recurrent paths are exactly the fixed paths, these are exactly the primitive paths, and there are Cat_{n-1} of them.*
- (ii) *If P has k primitive factors, then*

$$(5) \quad \tau(P) = k - 1.$$

The maximum entry time is $n - 1$, attained uniquely by $(UD)^n$.

- (iii) *For $1 \leq k \leq n$, the number of paths at depth $k - 1$ is*

$$(6) \quad |\{P \in \mathcal{D}_n : \tau(P) = k - 1\}| = \frac{k}{2n - k} \binom{2n - k}{n}.$$

- (iv) *Let $T = UQD$ be fixed, and factor its interior as $Q = Q_1 \cdots Q_r$ into primitive paths. For every $0 \leq d \leq r$, the unique basin state of T at depth d is*

$$(7) \quad P_d = (UQ_1 \cdots Q_{r-d} D) Q_{r-d+1} \cdots Q_r,$$

with the empty-prefix and empty-suffix conventions. Consequently

$$(8) \quad \mathcal{B}_T(u) = 1 + u + \cdots + u^r.$$

The unique terminal fibre of maximum size is the fibre over

$$(9) \quad T_n^{\max} = U(UD)^{n-1} D,$$

and it contains n states.

3. THE FACTOR-COUNT CLOCK

The first primitive component absorbs exactly one following component per round. The next lemma makes that statement precise at every time.

Lemma 3.1 (Closed orbit). *Let $P = C_1 \cdots C_k$, where $C_1 = UAD$. For every $0 \leq t \leq k - 1$,*

$$(10) \quad \Phi^t(P) = UAC_2 \cdots C_{t+1}DC_{t+2} \cdots C_k.$$

When $t = 0$, the block $C_2 \cdots C_{t+1}$ is empty; when $t = k - 1$, the suffix $C_{t+2} \cdots C_k$ is empty.

Proof. At $t = 0$, (10) reads $UADC_2 \cdots C_k = P$. Suppose the formula holds for some $t < k - 1$. The word

$$UAC_2 \cdots C_{t+1}D$$

is primitive: after its initial U , the Dyck word $AC_2 \cdots C_{t+1}$ stays at or above height one, and the displayed D is its first return to height zero. Hence the primitive factorisation of the right-hand side of (10) consists of that first word, followed by $C_{t+2}, \dots, C_k$. Applying (2) moves the closing D across C_{t+2} and gives

$$UAC_2 \cdots C_{t+1}C_{t+2}DC_{t+3} \cdots C_k,$$

which is (10) at time $t + 1$. Induction proves the claim. $\square$

Corollary 3.2 (Recurrent set and pointwise clock). *The statements in theorem 2.1(i)–(ii) hold.*

Proof. For $t < k - 1$, the factorisation displayed in the proof of lemma 3.1 has one first component and $k - t - 1$ untouched components, for a total of $k - t$ primitive factors. Thus each nonfixed step decreases κ by one. At $t = k - 1$, the formula is the single primitive path

$$(11) \quad E(P) = UAC_2 \cdots C_kD.$$

It is fixed, so $\tau(P) = k - 1$.

If $k = 1$, the definition fixes P . If $k > 1$, the strict decrease of κ prevents P from lying on a cycle. Fixed and recurrent paths are therefore exactly the primitive paths. Deleting the first U and last D is a bijection from primitive paths in $\mathcal{D}_n$ to $\mathcal{D}_{n-1}$, giving Cat_{n-1} fixed paths.

Every primitive factor has semilength at least one, so $k \leq n$. Equality holds only when all k factors have semilength one, and the only such factor is UD . In view of (5), the depth is at most $n - 1$, with equality exactly at $(UD)^n$. $\square$

The proof gives more than a global bound: the component statistic is the exact amount of time remaining at every state. In particular, no transient can merge with another transient of a different factor count before their depths agree.

4. COMPLETE TEMPORAL LAYERS

The clock reduces temporal enumeration to a component census. We include the coefficient calculation to fix the boundary case and normalisation, while treating the enumerative machinery as background.

Let

$$C(z) = \sum_{m \geq 0} \text{Cat}_m z^m$$

be the Catalan series, which satisfies $C(z) = 1 + zC(z)^2$. A primitive path is UAD with A an arbitrary Dyck path, so the semilength generating function for one primitive component is

$$(12) \quad R(z) = zC(z).$$

Proposition 4.1 (Ballot layer formula). *For $1 \leq k \leq n$, the number of paths in $\mathcal{D}_n$ with exactly k primitive factors is the right-hand side of (6).*

Proof. A path with k primitive factors is an ordered sequence of k primitive paths. By (12), its number is

$$(13) \quad [z^n]R(z)^k = [z^{n-k}]C(z)^k.$$

Put $m = n - k$ and $W = C - 1$. Then $W = z(1 + W)^2$. If $m = 0$, the coefficient in (13) is 1, which agrees with $k(2n - k)^{-1} \binom{2n-k}{n} = 1$ because $n = k$.

For $m \geq 1$, Lagrange inversion gives

$$\begin{aligned} [z^m]C(z)^k &= [z^m](1 + W)^k \\ &= \frac{1}{m} [w^{m-1}] k(1 + w)^{k-1} (1 + w)^{2m} \\ &= \frac{k}{m} \binom{2m + k - 1}{m - 1} \end{aligned}$$

$$= \frac{k}{2m+k} \binom{2m+k}{m}.$$

Substituting $m = n - k$ and using binomial symmetry yields

$$\frac{k}{2n-k} \binom{2n-k}{n-k} = \frac{k}{2n-k} \binom{2n-k}{n},$$

as required. $\square$

Combining [corollary 3.2](#) and [proposition 4.1](#) proves [theorem 2.1\(iii\)](#). It also gives the complete temporal polynomial

$$(14) \quad \sum_{P \in \mathcal{D}_n} u^{\tau(P)} = \sum_{k=1}^n \frac{k}{2n-k} \binom{2n-k}{n} u^{k-1}.$$

Across all semilengths, the same sequence construction yields

$$(15) \quad \sum_{n \geq 1} \sum_{P \in \mathcal{D}_n} u^{\tau(P)} z^n = \frac{R(z)}{1 - uR(z)} = \frac{zC(z)}{1 - uzC(z)}.$$

Equations (14)–(15) package the clock after it has been proved; generic generating-function extraction is not part of the residual theorem claim.

5. THE TERMINAL DEPTH-FIBRE ATLAS

The endpoint formula (11) absorbs every component after the first into the interior of one primitive path. Inverting it amounts to choosing how many final primitive components of the target interior should be moved back outside.

Theorem 5.1 (One source at every feasible depth). *Let $T = UQD \in \mathcal{D}_n$ be primitive, and let $Q = Q_1 \cdots Q_r$ be the primitive factorisation of its interior. For each $0 \leq d \leq r$, the path P_d in (7) is the unique path satisfying*

$$E(P_d) = T, \quad \tau(P_d) = d.$$

There is no basin state of T at any other depth.

Proof. Fix $d \in \{0, \dots, r\}$. The word

$$UQ_1 \cdots Q_{r-d}D$$

is primitive because its interior is a Dyck path. The suffix $Q_{r-d+1}, \dots, Q_r$ consists of d primitive paths. Hence P_d has exactly $d + 1$ primitive factors. By [corollary 3.2](#), its depth is d , and the endpoint formula gives

$$E(P_d) = UQ_1 \cdots Q_{r-d}Q_{r-d+1} \cdots Q_rD = T.$$

This proves existence.

For uniqueness, take any P with $E(P) = T$ and $\tau(P) = d$. The clock implies that P has $d + 1$ primitive factors, say

$$P = B_1B_2 \cdots B_{d+1}, \quad B_1 = UAD.$$

Its endpoint is $UAB_2 \cdots B_{d+1}D$. Equality with UQD implies

$$(16) \quad Q = AB_2 \cdots B_{d+1}.$$

Because A is a Dyck path, the boundary immediately after A is a return of Q unless A is empty. Each B_i for $i \geq 2$ is primitive. Therefore the unique factorisation of Q into primitive paths forces

$$A = Q_1 \cdots Q_{r-d}, \quad (B_2, \dots, B_{d+1}) = (Q_{r-d+1}, \dots, Q_r).$$

Thus $P = P_d$. The same argument shows that d must lie between 0 and r , so no other depth occurs. $\square$

The boundary cases are included in the construction. If $r = 0$, then $T = UD$ and only $d = 0$ occurs. For arbitrary r , the case $d = 0$ gives $P_0 = T$, while $d = r$ gives $P_r = (UD)Q_1 \cdots Q_r$.

In the ordered-tree language of (3), the fixed target has a root with one child S , and $Q_1, \dots, Q_r$ encode the ordered children of S . The source P_d is obtained by lifting the last d children of S , in order, to become root-level siblings after S . Hence the suffix-cut and suffix-lift descriptions are the same elementary inverse operation; the graft/lift representation itself receives zero contribution credit.

Corollary 5.2 (Fibre polynomial and unique maximum). *Equations (8)–(9) hold.*

Proof. By [theorem 5.1](#), the coefficient of u^d in $\mathcal{B}_T(u)$ is one for $0 \leq d \leq r$ and zero otherwise. This proves (8); in particular, the ordinary fibre size is $\mathcal{B}_T(1) = r + 1$.

The interior Q has semilength $n - 1$. Since every Q_i has positive semilength, $r \leq n - 1$, and hence $\mathcal{B}_T(1) \leq n$. Equality forces all $r = n - 1$ factors to have semilength one. The only primitive factor of semilength one is UD , so equality occurs exactly when $Q = (UD)^{n-1}$. This gives the unique target (9). $\square$

The terminal atlas is pointwise: it identifies the source word, not merely the number of sources. It also records when each source enters the fixed set, rather than collapsing all depths into one ordinary fibre count.

6. EXACT FINITE CONTROLS

The standard-library program `verify_p144.py` supplies finite counterexample pressure. It generates every Dyck path of semilength at most twelve. The literal map is implemented from the first two return positions, whereas the closed iterate is implemented separately from the initial primitive-factor list. Basin profiles are accumulated from the enumerated functional graph and compared with the constructive source formula for every fixed target and every feasible depth.

TABLE 1. Selected exact control sizes. The audit is exhaustive within each displayed semilength and uses no random sampling.

n	states	fixed targets	maximum depth	maximum fibre	assertions
1	1	1	0	1	24
4	14	5	3	4	277
8	1,430	429	7	8	28,901
12	208,012	58,786	11	12	4,308,839

Across semilengths 1 through 12, the run covers 290,511 states and 82,500 fixed targets. It performs 6,005,502 exact assertions and ends in `STATUS=PASS`. The frozen transcript is `verification_output.txt`. The audit checks closure, factor drop, every iterate, every temporal layer, the unique sharp source, all terminal depth profiles, and the unique maximum target.

Finite enumeration does not prove an all-parameter identity. The proofs in [sections 3 to 5](#) establish the theorem; the program can only reveal errors within its exhaustive range. It also supplies no evidence about priority or ownership.

7. OWNERSHIP BOUNDARY AND CONCLUSION

The ownership subtraction is now explicit. Huang–Tamari and the m -Tamari path literature own the lattice and atomic cover background [\[1, 3\]](#). Pallo’s right-arm rotations and Chapoton’s Dyck-path comparison own the comb order and its height-zero cover characterisation [\[2, 4\]](#). Pallo’s later leftmost left-rotation supplies a direct precedent for a unique deterministic leftmost-rotation scheduler, a rooted rotation tree, a grading, and a distance [\[5\]](#). Its unique terminal root separates that map from Φ_n , whose terminal set has size Cat_{n-1} , but the general leftmost-scheduling idea is not residual here.

Prime factorisation and component enumeration are also direct background [\[6\]](#). The standard contour bijection [\[7\]](#) turns the literal move into the root-child graft [\(3\)](#), makes the clock root degree minus one, and turns the inverse formula into a suffix lift. We therefore assign zero separate credit to the scheduler, atomic or ground-level rotations, primitive decomposition, comb/height-zero correspondence, root-degree clock, graft/lift representation, ballot layers, Catalan enumeration, and generic coefficient extraction.

After those deductions, the only retained internal residual is the exact conjunction, for the particular map [\(2\)](#), of the all-time iterate formula with the targetwise assertion that every feasible depth has one specified preimage. Its polynomial and extremal consequences are kept only as consequences of that conjunction. This is a bookkeeping residual, not a novelty or ownership claim: the bounded searches recorded in the source ledger did not establish an owner, but a non-hit provides no priority evidence. We do not classify other schedulers or arbitrary chains. The manuscript therefore remains owner-thin and anonymous under `HOLD_EXTERNAL`.

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